NatureClean AI

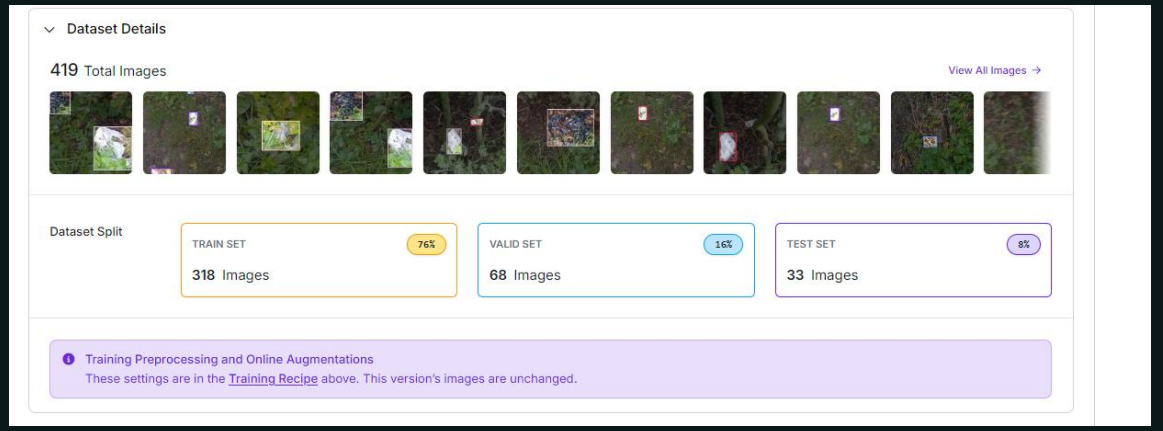
A photo. A checked count. A cleanup recorded.

AI for Good / Hackathon 7
Roboflow computer vision / SDG 15: Life on Land
Rayna Jong
Arber Deda

The local problem

Litter among plants, paths and soil

Our own photos and video frames from The Hague showed packaging, cans, bottles and other waste.



Our focus: helping existing volunteers log what they collect.

Who we built it for

Local cleanup volunteers in The Hague

A phone or computer, internet, and a cleanup photo

A quick first count, with the option to correct it

Organisers can follow participation through rankings

Outside our scope: fines, scientific litter surveys and verified impact claims.

The cleanup flow

01 Sign in and upload or take a cleanup photo

02 Roboflow detects objects and suggests a count

03 Volunteer checks, corrects and confirms

04 Supabase saves the counts and awards points

05 Dashboard, profile and leaderboard update

Our Roboflow model

419

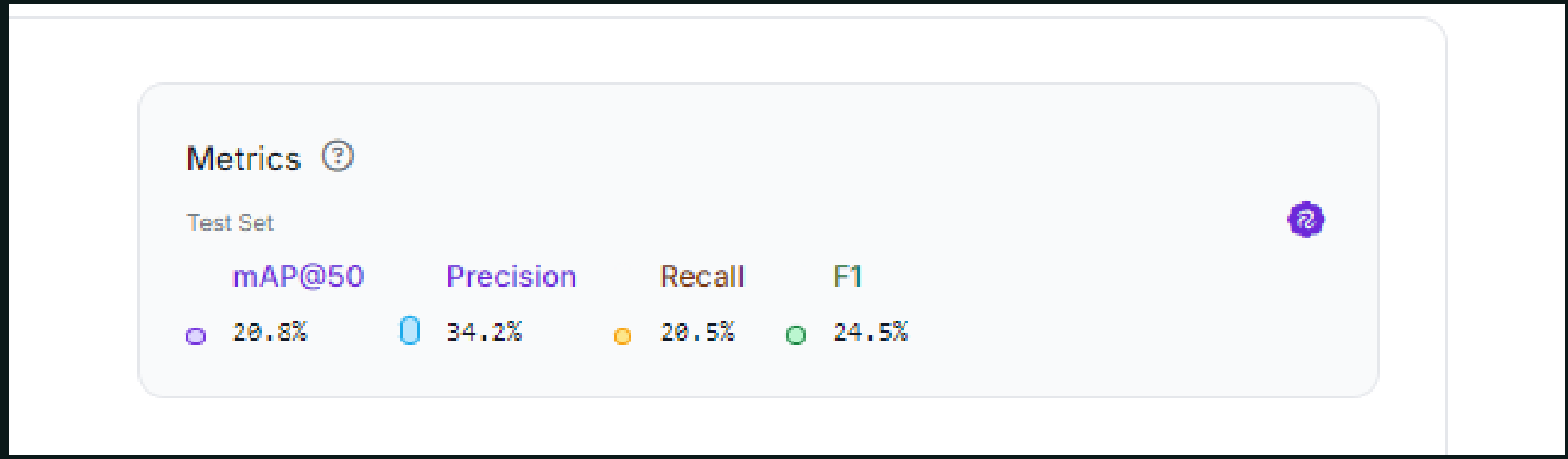
self-collected, labelled images

318 training 68 validation 33 test

RF-DETR Small / 7 trained litter classes
512 × 512 fit resizing / horizontal flip augmentation

Limitation: similar video frames and limited scene diversity.

What the results actually show



Many missed objects. Some wrong predictions.

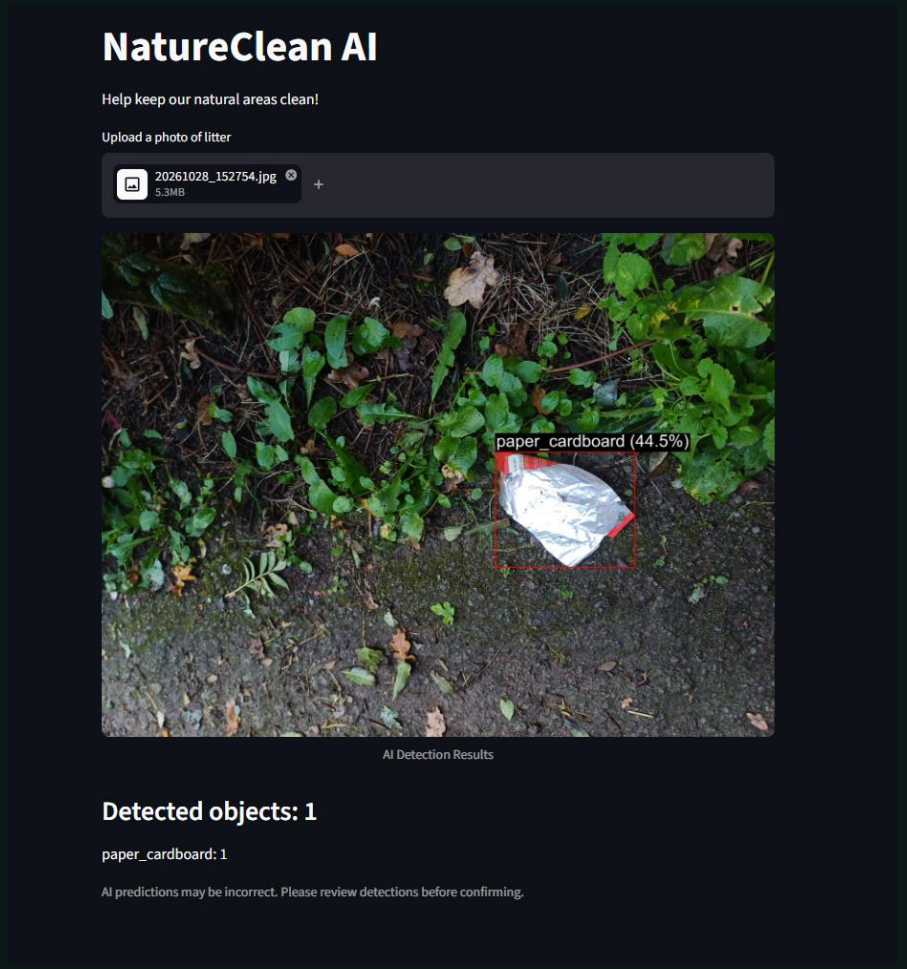
The model provides a first suggestion. A person checks the final count.

A new photo, an imperfect prediction

One object detected
44.5% confidence

The paper/cardboard label
is questionable for this packaging.

A detection can still need correction.



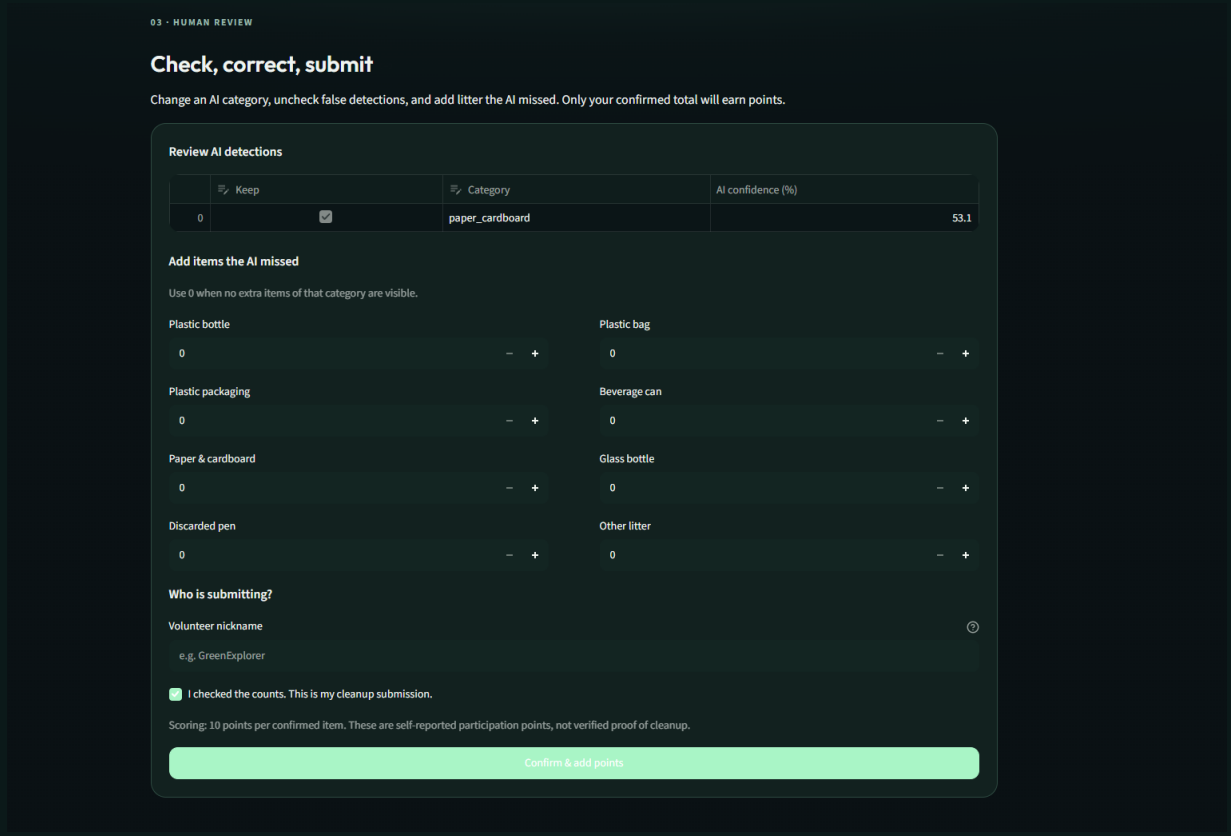
Earlier Streamlit build

The volunteer has the final say

Keep, remove or reclassify
Add anything the model missed
Confirm before saving

10 points per confirmed item

Points represent self-reported participation.



Development interface

Why a photo-based app?

A photo gives the volunteer a starting count

Saved history makes repeated participation visible

SDG 15: support litter removal in local green spaces

Still to test: is this easier than manual entry, and do points help people return?

Risks and safeguards

Missed litter can mean unfair suggested scores

Correction before scoring. Broader data and phone testing next.

Photos can contain personal information

Avoid faces. Use public nicknames. Review cloud retention before rollout.

Points can reward exaggerated submissions

Block identical images. Keep prizes inactive until verification improves.

Live demo and next steps

Sign in, upload a new photo and run detection

Review the result, confirm and check the updated points

Open the leaderboard and try the duplicate-photo check

Next: more varied data and a small volunteer pilot.

Portfolio evidence still needed: final V2 submission and updated leaderboard.